

Differentiability of real functions

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Calculus

Differentiable functions

- **Definition.** Let $I \subseteq \mathbb{R}$ be a nonempty open interval, $f : I \rightarrow \mathbb{R}$ and $x \in I$. We say that f is differentiable at the point x if the limit

$$\lim_{y \rightarrow x} \frac{f(y) - f(x)}{y - x}$$

exists and finite. In this case the above limit is denoted by $f'(x)$ and is called the derivative of f at the point x . If f is differentiable at any point of I then we say that f is differentiable on I . The function f' itself is called the first order derivative of the function.

- **Remarks.**

- ★ If f is differentiable at x then it is continuous at x .
- ★ The absolute value function is not differentiable at the point 0. The one sided limits of the function $y \mapsto \frac{|y|}{y}$ at zero are finite, but different.
- ★ Geometrically, $f'(x)$ is nothing else but the slope of the tangent line touching the graph of f at the point $(x, f(x))$.

Derivatives of elementary functions

Theorem. The power functions, exponential functions, logarithm functions, trigonometric functions and hyperbolic functions are differentiable on their domain, furthermore, for their derivatives, we have the following formulas.

- If $k \in \mathbb{Q}$ and $f(x) := x^k$, then $f'(x) = kx^{k-1}$.
- If $a > 0$ and $f(x) := a^x$, then $f'(x) = \ln(a)a^x$. Particularly, if $a = e$, then $f'(x) = e^x$.
- If $a > 0$, $a \neq 1$ and $f(x) := \log_a(x)$, then $f'(x) = \frac{1}{\ln(a)x}$. Particularly, if $a = e$, then $f'(x) = \frac{1}{x}$.
- If $f(x) := \sin(x)$, then $f'(x) = \cos(x)$. If $f(x) := \cos(x)$, then $f'(x) = -\sin(x)$.
- If $f(x) := \sinh(x)$, then $f'(x) = \cosh(x)$. If $f(x) := \cosh(x)$, then $f'(x) = \sinh(x)$.

Differentiability and operations

Theorem. Let $I \subseteq \mathbb{R}$ be a nonempty open interval, $f, g : I \rightarrow \mathbb{R}$ and $x \in I$. If f and g are differentiable at the point x , then the following statements hold.

- ★ For all $\alpha, \beta \in \mathbb{R}$, the function $\alpha f + \beta g$ is differentiable at x , furthermore

$$(\alpha f + \beta g)'(x) = \alpha f'(x) + \beta g'(x).$$

- ★ The product $f \cdot g$ is differentiable at x , furthermore

$$(f \cdot g)'(x) = f'(x)g(x) + f(x)g'(x).$$

If, in addition, $g(x) \neq 0$, then the ratio $\frac{f}{g}$ is differentiable at x , furthermore

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}.$$

Differentiability of the composite function

Theorem. Let $I \subseteq \mathbb{R}$ be a nonempty open interval, $x \in I$, furthermore $f : I \rightarrow \mathbb{R}$ and $g : f(I) \rightarrow \mathbb{R}$ be functions such that f is differentiable at x and g is differentiable at $f(x)$. Then $g \circ f$ is differentiable at x and

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x).$$

• **Example.** Let $I := \mathbb{R}$, $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) := 2x + 1$ and $g : \mathbb{R} \rightarrow [-1, 1]$, $g(u) := \cos(u)$. According to the previous slide, the functions f and g are differentiable on their domain, hence, in view of the above theorem,

$$(g \circ f)(x) = \cos(2x + 1), \quad (x \in \mathbb{R})$$

is differentiable on its domain. By $f'(x) = 2$ and $g'(u) = -\sin(u)$, we have

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x) = -\sin(2x + 1) \cdot 2 = -2 \sin(2x + 1).$$

Differentiability of the inverse function

Theorem. Let $I \subseteq \mathbb{R}$ be a nonempty open interval and $f : I \rightarrow \mathbb{R}$ be a continuous, strictly monotone function. If f is differentiable at the point $x \in I$ and $f'(x) \neq 0$, then f^{-1} is differentiable at $f(x)$ and

$$(f^{-1})'(f(x)) = \frac{1}{f'(x)}.$$

Differentiability of the inverse function

• **Example 1.** In view of our list of derivatives of basic functions, the derivative of $f^{-1}(x) := \log_a(x)$, for some $a > 0$, $a \neq 1$, is $\frac{1}{\ln(a)x}$. Let us prove this statement using the differentiation formula of the inverse function. Keeping the above notations, $f(u) = a^u$, where $u = f^{-1}(x)$. Then we have $f'(u) = \ln(a)a^u$, which is different from zero on its domain. Thus

$$\begin{aligned}(f^{-1})'(x) &= (f^{-1})'(f(u)) = \frac{1}{f'(u)} = \frac{1}{\ln(a)a^u} \\ &= \frac{1}{\ln(a)a^{f^{-1}(x)}} = \frac{1}{\ln(a)a^{\log_a(x)}} = \frac{1}{\ln(a)x}.\end{aligned}$$

• **Example 2.** Let $f(u) := \tan(u)$. Let us calculate the derivative of the $f^{-1}(x) = \arctan(x)$ function. Now we have

$$\begin{aligned}(f^{-1})'(x) &= (f^{-1})'(f(u)) = \frac{1}{f'(u)} = \cos^2(u) = \cos^2(\arctan(x)) \\ &= \left(\frac{1}{\sqrt{1+x^2}} \right)^2 = \frac{1}{1+x^2}.\end{aligned}$$

Higher order derivatives

Definition. Let $I \subseteq \mathbb{R}$ be a nonempty open interval, $x \in I$ and $f : I \rightarrow \mathbb{R}$ be a function which is differentiable at x . Let $n \in \mathbb{N}$ with $n \geq 2$ and suppose we have already defined the $(n-1)$ -order derivative of f . We say that f is n -times differentiable at x if its $(n-1)$ -order derivative exists on an open neighborhood of x and the $(n-1)$ -order derivative is differentiable at the point x . The n th order derivative of f is denoted by $f^{(n)}$, whenever $n \geq 4$. Otherwise, we shall use the notations f'' or f''' if $n = 2$ or $n = 3$, respectively.

• **Examples.** Let $f(x) := 2x^5 + x^3 + 7x + 11$ and $g(x) := 3^x$. Then

$$f'(x) = 10x^4 + 3x^2 + 7, \quad f''(x) = 40x^3 + 6x,$$

$$f'''(x) = 120x^2 + 6, \quad f^{(4)}(x) = 240x, \quad f^{(5)}(x) = 240 \text{ and } f^{(n)}(x) = 0 \text{ if } n \geq 6,$$

furthermore, for all $n \in \mathbb{N}$, we have

$$g^{(n)}(x) = \ln^n(3) \cdot 3^x.$$

L'Hospital's Rule

Theorem. Let $I \subseteq \mathbb{R}$ be a nonempty open interval, $p \in \mathbb{R} \cup \{-\infty, +\infty\}$ be an accumulation point of I and $f, g : I \rightarrow \mathbb{R}$ be differentiable functions on I such that $0 \notin g(I)$. Assume that for the limits

$$A := \lim_{x \rightarrow p} f(x) \quad \text{and} \quad B := \lim_{x \rightarrow p} g(x)$$

we have $A, B \in \{0, -\infty, +\infty\}$. If the limit

$$\lim_{x \rightarrow p} \frac{f(x)}{g(x)}$$

exists and $g'(x) \neq 0$ whenever $x \in I$, then the limit

$$\lim_{x \rightarrow p} \frac{f'(x)}{g'(x)}$$

exists as well and

$$\lim_{x \rightarrow p} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow p} \frac{f(x)}{g(x)}.$$